Causal Inference & Machine Learning

Prediction, Causality, and Modern Methods

Kristian Boroz - opencampus.sh

Week02

Why Machine Learning in Economics?

- Explosion of high-dimensional data:
  - Text, images, transactions
  - User-level behavioral data
- Traditional econometrics struggles when $P \approx N$ or $P > N$
- ML provides:
  - Flexible functional forms
  - Automated model selection
  - Strong predictive performance

Key Concept

Core Question

When should we prioritize prediction vs causal inference?

- ML: focuses on prediction accuracy
- Econometrics: focuses on causal interpretation

Two Types of Machine Learning

Supervised Learning

- Independent observations
- Stable environment
- Estimate:

$$E[Y|X = x], \quad P(Y = y|X = x)$$

Unsupervised Learning

- No labels
- Discover structure in features

What is New About ML?

- Flexible, high-capacity models
- Ability to fit complex nonlinear relationships
- Data-driven model selection
- Regularization to avoid overfitting

Classification

- Goal: assign class labels
- Example:

$$P(Y = \text{cat}|X) = 0.95$$

- Neural networks learn features automatically

Ability to Fit Complex Shapes

- Traditional models:
  - Linear or low-dimensional nonlinear
- ML models:
  - Arbitrary nonlinear functions
  - High interaction terms

Prediction Objective

- Goal:

$$\mu(x) = E[Y|X = x]$$

- Minimize Mean Squared Error:

$$MSE = \mathbb{E}[(Y - \hat{\mu}(X))^2]$$

Key Assumptions

- Independent observations
- Same joint distribution in train/test

Important

Prediction works only in a **stable environment**

Train-Test Framework

- Train model on dataset
- Evaluate on hold-out test set
- Ground truth observed in test data

Cross-Validation

- Split into k folds (often $k = 10$)
- Train on $k - 1$ folds
- Validate on remaining fold
- Repeat and average errors

Quiz 1

Why is cross-validation used?

1. To increase bias
2. To select tuning parameters
3. To reduce sample size
4. To estimate causal effects

Quiz 1 Solution

1. To increase bias
2. To select tuning parameters
3. To reduce sample size
4. To estimate causal effects

Bias-Variance Tradeoff

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \sigma^2$$

- Complex models:
  - Low bias, high variance
- Simple models:
  - High bias, low variance

Overfitting

- Model captures noise
- Excellent training performance
- Poor generalization

Regularization

$$\min \sum (Y - \hat{Y})^2 + \lambda ||\beta||^2$$

- Penalizes model complexity
- Reduces variance

Quiz 2

What happens when λ increases?

1. Model becomes more complex
2. Model becomes simpler
3. Variance increases
4. Bias decreases

Quiz 2 Solution

1. Model becomes more complex
2. Model becomes simpler
3. Variance increases
4. Bias decreases

ML vs Econometrics

- ML:
  - Focus: prediction
  - Evaluated via test error
- Econometrics:
  - Focus: causal inference
  - Evaluated via identification

OLS vs ML

- OLS:
  - Optimal when $N >> P$
  - Minimizes in-sample error
- ML:
  - Works when $P > N$
  - Uses regularization

Identification Problem

$$Y = \beta X + \varepsilon$$

- If X correlated with ε:
  - $\Rightarrow$ biased estimates

Quiz 3

Why can't ML solve causality directly?

1. Too complex
2. Optimizes MSE, not causal parameters
3. Requires small datasets
4. Cannot handle nonlinearities

Quiz 3 Solution

1. Too complex
2. Optimizes MSE, not causal parameters
3. Requires small datasets
4. Cannot handle nonlinearities

Key Problem

Standard ML methods produce biased causal estimates

- Optimize prediction, not identification
- Confounding variables distort estimates

Solutions

- Sample splitting
- Cross-fitting
- Orthogonalization
- Double ML

Double Machine Learning

- Estimate nuisance parameters using ML
- Remove bias via orthogonalization
- Allows valid inference

CATE (Heterogeneous Effects)

- Estimate:

$$\tau(x) = E[Y(1) - Y(0)|X = x]$$

- Allows personalized treatment effects

Methods for CATE

- Causal Trees
- Random Forests
- X-Learners
- BART

Quiz 4

What is CATE?

1. Average treatment effect
2. Conditional treatment effect
3. Prediction error
4. Model variance

Quiz 4 Solution

1. Average treatment effect
2. Conditional treatment effect
3. Prediction error
4. Model variance

Policy Learning

- Goal: choose optimal treatment policy
- Based on predicted treatment effects

Contextual Bandits

- Sequential decision-making
- Trade-off:
  - Exploration vs exploitation

Applications in Economics

- Demand estimation
- Pricing
- Policy targeting
- Text analysis

Panel Data + ML

- Synthetic control methods
- Matrix completion
- Regularized regression

Structural Models

- Combine ML with economic theory
- Handle large-scale consumer data

Predictions

- Regularization becomes standard
- More predictive modeling
- Integration of ML + causal inference

Final Quiz

What is the main takeaway?

1. ML replaces econometrics
2. ML complements econometrics
3. Causality is no longer needed
4. OLS is obsolete

Final Quiz Solution

1. ML replaces econometrics
2. ML complements econometrics
3. Causality is no longer needed
4. OLS is obsolete

Conclusion

- ML: powerful prediction tool
- Econometrics: causal understanding
- Future: integration of both

The Secret Sauce

Three key ingredients of ML:

1. Flexible functional forms
2. Regularization (limit complexity)
3. Tuning (learn optimal complexity)

Core Idea

ML balances fit vs overfitting automatically

Prediction Problem (Formal)

Given:

- Data $(Y_i, X_i)_{i=1}^n$
- Loss function $\ell(\hat{y}, y)$

Goal:

$$\hat{f} = \arg\min_f \mathbb{E}[\ell(f(X), Y)]$$

- Expectation over new data
- Same distribution as training data

Squared Error Loss

For regression:

$$\ell(\hat{y}, y) = (\hat{y} - y)^2$$

$$L(f) = \mathbb{E}[(f(X) - Y)^2]$$

- Target:

$$f^*(x) = E[Y|X = x]$$

Classification Loss

Binary outcome $Y \in \{0, 1\}$

- Prediction $\hat{y} \in \{0, 1\}$
- Errors:
  - False positive (Type I)
  - False negative (Type II)

Trade-off

Different losses imply different optimal models

Standard Regression (OLS)

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{n} \sum (Y_i - \beta' X_i)^2$$

- Optimal under classical assumptions
- But:
  - Not robust in high dimensions
  - Overfits when P large

Bias-Variance Decomposition

$$E[(\hat{Y} - Y)^2] = Bias^2 + Variance + \sigma^2$$

- Bias: approximation error
- Variance: sensitivity to data
- σ^2: irreducible noise

Approximation vs Overfitting

As model complexity increases:

- Bias $\downarrow$ (better approximation)
- Variance $\uparrow$ (overfitting)

Trade-off

Optimal model balances both

Regularization Idea

Instead of:

$$\min \sum (Y - \hat{Y})^2$$

Use:

$$\min \sum (Y - \hat{Y})^2 + \lambda \cdot Penalty$$

- Controls complexity
- Prevents overfitting

LASSO

$$\min \sum (Y - \beta X)^2 + \lambda \sum |\beta_j|$$

- Produces sparse solutions
- Performs variable selection

Ridge Regression

$$\min \sum (Y - \beta X)^2 + \lambda \sum \beta_j^2$$

- Shrinks coefficients
- Keeps all variables

Quiz: LASSO vs Ridge

Which is correct?

1. LASSO keeps all variables
2. Ridge performs selection
3. LASSO can set coefficients to zero
4. Ridge creates sparse models

Solution

1. LASSO keeps all variables
2. Ridge performs selection
3. LASSO can set coefficients to zero
4. Ridge creates sparse models

Prediction vs Estimation

Prediction:

- Focus on $\hat{Y} = \hat{f}(X)$
- Goal: minimize prediction error

Estimation:

- Focus on β
- Goal: causal interpretation

Key Insight

Good prediction does NOT imply good estimation

- Many models predict equally well
- But imply very different parameters

Why ML Fails for Estimation

- Regularization introduces bias
- Model selection introduces instability
- High-dimensional correlations distort coefficients

LASSO Biases

Selection biases:

- Important variables dropped
- Irrelevant variables included

Shrinkage bias:

- Coefficients biased toward zero

Types of Bias (Important)

- Omitted variable bias
- Shrinkage bias
- Expansion bias
- Compactification bias

Key Message

Bias is unavoidable in regularized models

Ridge Bias

- All coefficients shrunk toward zero
- Correlated variables distort estimates

Trees: Instability

- Small data changes $\rightarrow$ different splits
- Leads to unstable estimates

Solution

Use ensembles (random forests)

Prediction vs Coefficients

- $\hat{Y}$ can be accurate
- $\hat{\beta}$ can be:
  - Biased
  - Inconsistent
  - Unstable

Quiz

Why is ML good at prediction?

1. It estimates true parameters
2. It minimizes prediction loss
3. It avoids bias
4. It ensures consistency

Solution

1. It estimates true parameters
2. It minimizes prediction loss
3. It avoids bias
4. It ensures consistency

Key Takeaways

- ML delivers strong predictions
- Does NOT guarantee causal interpretation
- Coefficients often not meaningful

Applications of Prediction Tools

1. Data preprocessing
2. Prediction tasks
3. Input into causal models